Alpha Technology Sales & Revenue Analysis

Problem Statement -:

The client Alpha Technology lacked a clear understanding with their sales performance and revenue generation. Suffering loses in some months, it was unclear which factors caused it. With good quality data in hand, Alpha Technology needed a deep analysis and visualisation for their data to getter better insights in terms of profitability, loss, quantity ordered and required, and the various factors with respect to customers belonging to different states, payment mode, category, sub-category etc.

Objective -:

The objective of this analysis and visualisation is to study the sales and revenue generated and to interpret customer behaviour from all over India depending on factors such as city, state, category, sub-category, quantity, the payment-mode being used.

This is to be ready with a plan to counter any loss expected in particular months or the expected quantity to have and the popularity of any particular sub-category among the customers.

Purpose / Client requirements - :

The client had two sets of good quality data, one for customers and one for orders. The customers data contained information helping us to identify their cities, states, dates on which they made the purchases as well as to identify the order they made from the orders data which contains the information for individual orders, telling us the category and sub-category, quantity, amount, profit (positive or negative), and mode of payment used by the customers.

The client required to get better insights on sales and revenue they generated for the year, whether they made profit or suffered any loss for any particular month. Some other requirements included:

a. Top states and sub-category in which they made the maximum profit.
b. States that made the highest profit.
c. Payment mode preferred by the customers.
d. Customers spending the most.
e. Category selling the most.
f. Miscellaneous aggregated data.

Dashboard-:

1. Power BI, a business intelligence and data visualisation tool was used to make the required charts. The two datasets described above, customers and orders were loaded onto the memory. The data was examined initially to understand the contents of it. Visualizations section (Chart pane) provides with more than 20 options for making graphs.

2. **Profit by Month** was created using the Stacked Column Chart option from the visualisation pane, X-axis representing the months and Y-axis the summation of profit for that particular month. With the help of 'format visual' a rule was applied to the columns resulting the loss months to be observed in red and profit in blue.

3. **Profit by State, Profit by Sub-category, and Top Customers** chart were created using the Stacked Bar Chart option. Additionally, only the top bars were filtered out using the Filter option and individual colours were assigned to the bars.

4. Two Donut Charts were created **PaymentMode** and **Top Selling Category.** The data selected for PaymentMode chart were PaymentMode column and ∑ quantity from the Orders dataset. For Top Selling Category, the columns chosen were Category and ∑ quantity.

5. Four sets of data are displayed in the form of cards using the Data Cards chart from the chart pane. It gives us following information: **Sum of Amount, Sum of Profit, Sum of OrderAverage, Sum of Quantity**. Using the DAX functionality from Power BI, a new column was added to calculate Sum of OrderAverage.

6. Another important feature from the chart pane we used was the **Slicer** which helped us visualising the data quarterly as well as state wise.

7. Furthermore, a Map chart named **State and City** representing information in the form of dots for city and states can be seen on the dashboard.

Each chart was modified and made to fit on the dashboard according to its significance and importance with respect to our requirements. Extra care was taken by looking at minute details such as transparency, borders, and titles. Major part of our dashboard represents charts that are aggregated on amount and profit.

The charts are quite simple to read, understand and interpret. One doesn't need to be fully proficient in Power BI to use the dashboard. By simply hovering over, let's say the Profit by Month chart any particular month can be selected by clicking on the bar representing it and the data for specific month gets highlighted not only for the Profit by Month but also for other charts on the board. Slicers as mentioned before can give us data quarterly, by clicking on any one of the four quarter options (Qtr1, Qtr2, Qtr3, Qtr4), the data on all the other charts changes for that respective quarter only. The other slicer is helpful for segregating the data state-wise, by clicking on any one of the states from the dropdown menu, the entire data on all graphs represents for that state alone.

Solved Problems -:

1. The first and foremost problem was to have a basic understanding of sales and revenue performance for the whole year which our dashboard successfully solves it. The Profit by Month chart conveys that the November month was the highest profit-making month and Alpha Technology failed to understand which part of year suffered loses which can be observed by the graph, it being May, July, September, December.

2. The client needed a deeper understanding to customer's behaviour which was achieved with the help of Top Customers, Top Selling Category, Profit by Sub-Category, and Profit by State graphs. It could be easily grasped that states that made the most profit were Maharashtra, Madhya Pradesh, Uttar Pradesh conveying that selling in these states is more profitable compared to the rest.

3. Top Customers graph displaying Rajesh, Abhishekh, Vijay etc. is an interactive chart which means that clicking on these names, we understand the cities and states they belong to from the Map (State and City) chart present on top left part of the dashboard. Rajesh belonging from New Delhi and Abhishekh from Mumbai, it gives an estimate to the client about customers who would purchase the most and make them the highest profit.

4. Top Selling Category and Profit by Sub-Category charts show us the parent category in which the highest purchases were done (clothing) and the sub-category that made the highest profit i.e. printers and bookcases.

5. Another important aspect of customer behaviour could be observed with the help of PaymentMode chart. Cash on Delivery (COD) was the highest chosen mode of payment preferred by customers with 43.47% of the total payment made and the least being EMI 10.49%.

6. Another requirement for displaying aggregated data such as the overall total amount, total profit was also fulfilled as observed by four data cards, Sum of Amount, Sum of Profit, Sum of OrderAverage, and Sum of Quantity from the dashboard.

7. Apart from client's request, one another important feature was added on the dashboard in the form of two slicers. The first slicer is the quarter slicer i.e. has 4 buttons for each quarter of the year. Clicking on any of these buttons modifies all the charts present on the dashboard with respect to that quarter only. Second slicer is the

state slicer which has a dropdown menu for all the states and clicking on any one of the state again modifies all the graphs present on the dashboard for that state alone.

Conclusion -:

It can be concluded that Power BI proves to be a very important tool when it comes to analysing and visualising data for understanding data trends and patterns which is not easily feasible when it comes to do it from raw data in the form of tables by a human.

Alpha Technology wanted a deeper analysis and interpretation for the data in-hand and the dashboard created with the help of Power BI is able to satisfy all their requirements with some additional features.

It not only helps to visualise in the form of charts but also to make the graphs interactive giving additional information about each element present on the dashboard. Hence, in this way a deeper analysis, transformation and visualisation was successfully performed for the data provided by Alpha Technology.

Dashboard Photo -:

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